

SQL Practice – Instagram Clone Database

Complete MySQL practice solution for database creation, test data, SELECT queries, and RDBMS vs NoSQL comparison.

1. Create the Database

```
CREATE DATABASE insta_clone_practice;
```

```
USE insta_clone_practice;
```

2. Create the users Table

```
CREATE TABLE users (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    username VARCHAR(100),  
    email VARCHAR(150),  
    created_at DATETIME  
);
```

3. Insert 3 Sample Users

```
INSERT INTO users (username, email, created_at)  
VALUES  
( 'jinal123', 'jinal@gmail.com', '2026-09-22 10:00:00'),  
( 'rahul_dev', 'rahul@yahoo.com', '2026-09-22 11:30:00'),  
( 'priya_test', 'priya@gmail.com', '2026-09-22 12:15:00');
```

```
SELECT * FROM users;
```

Expected Result

id	username	email	created_at
1	jinal123	jinal@gmail.com	2026-09-22 10:00:00
2	rahul_dev	rahul@yahoo.com	2026-09-22 11:30:00
3	priya_test	priya@gmail.com	2026-09-22 12:15:00

4. Find Gmail Users

```
SELECT *  
FROM users  
WHERE email LIKE '%gmail.com';
```

Explanation: The % wildcard represents any characters before **gmail.com**. The query therefore returns users whose email address ends with gmail.com.

5. RDBMS vs NoSQL Comparison

Difference	RDBMS – MySQL	NoSQL – MongoDB	Real-world example
Data structure	Tables, rows and columns	Flexible documents and collections	Instagram user data can be structured in MySQL tables or stored as JSON documents in MongoDB
Relationships	Uses relationships and SQL JOINs	Often uses embedded documents or references	Users and posts can be connected using foreign keys in a relational database, while posts can be embedded in user documents
Schema	Usually predefined/fixed schema	Flexible schema; documents can have different fields	WhatsApp message documents can contain different metadata depending on the type of message

Complete SQL Script

```
CREATE DATABASE insta_clone_practice;
```

```
USE insta_clone_practice;
```

```
CREATE TABLE users (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    username VARCHAR(100),  
    email VARCHAR(150),  
    created_at DATETIME  
);  
  
INSERT INTO users (username, email, created_at)  
VALUES  
('jinal123', 'jinal@gmail.com', '2026-09-22 10:00:00'),  
('rahul_dev', 'rahul@yahoo.com', '2026-09-22 11:30:00'),  
('priya_test', 'priya@gmail.com', '2026-09-22 12:15:00');  
  
SELECT * FROM users;  
  
SELECT *  
FROM users  
WHERE email LIKE '%gmail.com';
```

Testing Practice Covered: Database creation, table creation, primary key, auto-increment, test-data insertion, SELECT, LIKE operator, wildcard %, and basic RDBMS vs NoSQL concepts.